Homework 1
Statistics 2103

Some of the following exercises are taken from *Probability and Statistical Inference* by Hogg and Tanis.

1. Let X be a random variable with probability mass function $P(X = x) = \frac{x+2}{15}$ for x= 0, 1, 2, 4.
   (a) What is $\mathbb{E}(X)$?
   (b) What is $\text{var}(X)$?
   (c) Define a new random variable $Y = 2X - 1$. What is the covariance of X and Y?

2. Let X be a random variable with expected value $\mathbb{E}(X) = \mu$ and variance $\text{var}(X) = \sigma^2$.
   (a) What is $\mathbb{E}\left(\frac{X-\mu}{\sigma}\right)$?
   (b) What is $\text{var}\left(\frac{X-\mu}{\sigma}\right)$?

3. Let W be the number of tries it takes to plug in a USB correctly. The probability distribution of W is

w	1	2	3
P(W=w)	0.45	0.50	0.05

   Let Y be a binary random variable, which equals 1 if the USB is successfully plugged in on the first try, and 0 otherwise.
   (a) Write out the joint distribution of W and Y in a table of the form:

	W=1	W=2	W=3
Y=1			
Y=0			

   (b) Given that you did not plug in the USB correctly on the first try, what is the probability that you will not plug it in correctly on the second try either? In other words, determine $P(W = 3|Y = 0)$.

4. Suppose you are driving to a doctor's appointment, but you are running late. You can take one route that has six stoplights, or another, longer route that has two stoplights. If you stop at more than half of the lights on either route, you will be late for the appointment. Assume independence, and assume that the chance of stopping at each light is $p = 0.5$. Which route should you take?

5. Let M be a random variable representing a student's score on the math section of the GRE exam, and V be a random variable representing their score on the verbal section. Suppose the distributions of M and V have the following characteristics:

$$\begin{array}{cc} \mathbb{E}(M) = 158 & \text{var}(M) = 10^2 \\ \mathbb{E}(V) = 151 & \text{var}(V) = 8^2 \\ \mathbb{E}(M+V) = 309 & \text{var}(M+V) = 15^2 \end{array}$$

   (a) What is the value of the correlation between the random variables M and V?

   (b) The file `scores.csv` contains the math and verbal scores for 1000 randomly selected students. For each student, calculate the difference between their math and verbal scores, $d_i = M_i - V_i$

      i. Make a histogram of the calculated differences d_i.

      ii. Calculate the sample mean and sample standard deviation of the differences.

      iii. How does the sample standard deviation of the differences compare to the standard deviations of the distributions of M and V?